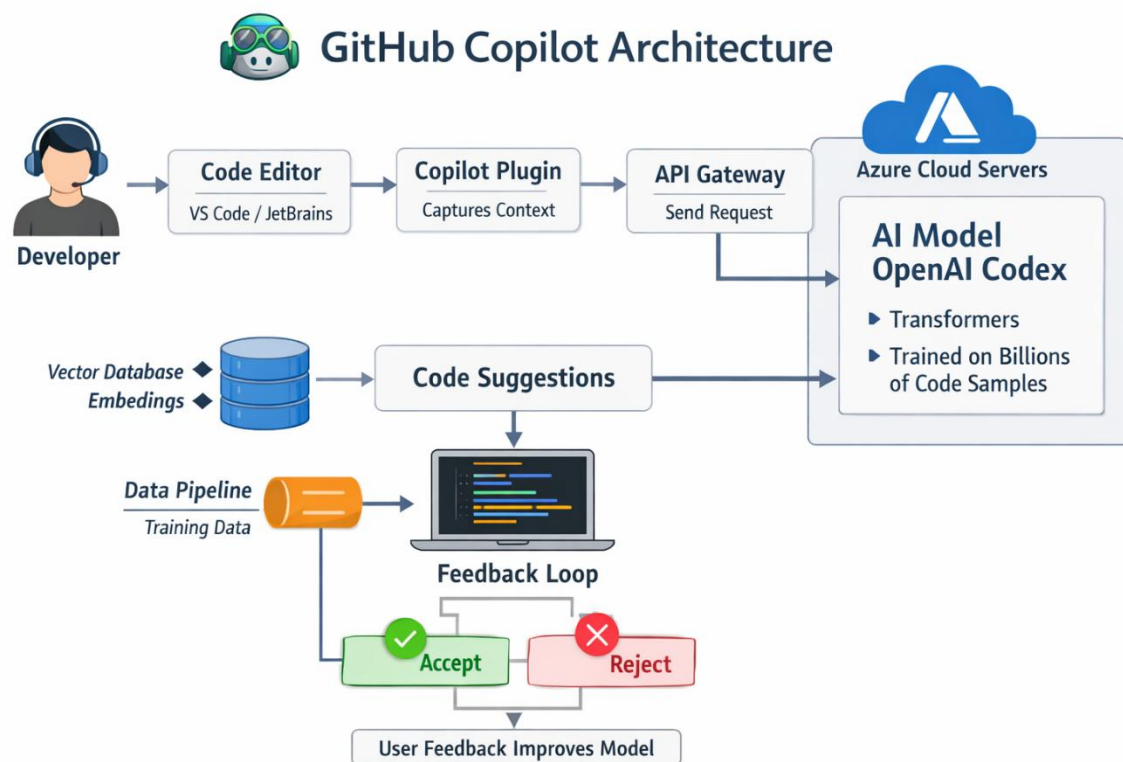


Research Assignment

GitHub Copilot :

System Architecture Diagram:



Technology Stack

- A **Technology Stack** is the **collection of tools, frameworks, programming languages, and platforms** used to build and run a software or AI system. Think of it as the **ingredients** that make the system work — each layer has a specific role.

Layer	Role	Examples
Frontend	Where developers interact with Copilot.	VS Code, JetBrains Plugin

Backend	Handles requests and connects to servers.	<i>Azure Cloud, API Gateway</i>
AI Frameworks	Build and train the AI model.	<i>PyTorch, Transformers</i>
Model Layer	The brain that predicts code.	OpenAI Codex
Data Infrastructure	Stores and manages training data	Embeddings, Vector Databases, Data Pipelines
Deployment	Where the system runs.	Azure Cloud AI

DATA FLOW

Developer Writing Code

- The developer types in a **code editor** like *VS Code* or *JetBrains*.
- Copilot observes the **context** — what language, function, and comments are being written.

Copilot Plugin Captures Context

- The plugin sends this context to the **API Gateway**.
- It acts as a bridge between your computer and the **cloud servers**.

API Gateway → Azure Cloud Servers

- The request travels to **Azure Cloud**, where the **OpenAI Codex model** lives.
- Codex is built using **Transformers** and trained on **billions of code samples** from public repositories.

AI Model (OpenAI Codex)

- The model processes your input and generates **code suggestions**.
- It uses **embeddings** (numerical representations of code) stored in **vector databases**.

Code Suggestions

- The AI sends back possible completions to your editor.
- You see suggestions instantly while typing.

Feedback Loop

- You can **Accept** ☒ or **Reject** ✖ suggestions.
- This feedback helps improve the model's accuracy over time.

How it runs?

- GitHub Copilot runs on **Cloud AI** (Microsoft Azure).
- It doesn't process data locally — instead, it sends requests to the cloud for **real-time inference**.
- This allows **scalability**, **speed**, and **continuous learning** from millions of developers worldwide.
- **Developer → Plugin → API → Codex Model → Suggestion → Feedback**

We use AI frame works like PyTorch, Transformers, Codex , Data infrastructure like Embeddings, Vector Databases, Data Pipelines , system Design like APIs, Inference Flow, Feedback Loop , cloud AI like Azure Cloud for scalability and updates